

Major League Baseball's First Year Player Draft: A Natural Laboratory for the Study of Bargaining

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**Christopher Garmon¹****Abstract**

Major League Baseball's (MLB) Draft is a natural laboratory for the study of bargaining: There are no bonus restrictions, draft picks cannot be traded, and there are no commitment issues (i.e., players "declaring" for the draft). This article tests the implications of bargaining theory using data from the 2003-2010 MLB drafts. In 2007, new draft rules were introduced, but not uniformly across the draft, making it possible to estimate the effect of each change. Although the rule changes were designed to lower bonuses, they had the opposite effect. The rule changes led to \$178 million in extra compensation to drafted players.

Keywords

bargaining, baseball, draft, framing effects

Introduction

The First Year Player Draft,¹ conducted each year in early June, is Major League Baseball's (MLB) primary means of acquiring new talent. Over 75% of the players currently on MLB rosters entered professional baseball through the draft.² Like the drafts in most team sports, MLB's draft can be thought of as an agreement between

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teams to divide the amateur labor market. Each team is granted the exclusive right to negotiate an initial labor contract with each of its draft picks and, if successful in initially signing the player, maintains this exclusive right over the player's services until the player accumulates 6 years of major league experience. Each team and drafted player are given a fixed time horizon to complete the initial contract and, if no agreement is reached, the team cannot select the player in subsequent drafts.³ Unlike the National Football League (NFL) and National Basketball Association (NBA), MLB teams cannot trade draft picks, nor can they trade drafted players until a year after they are drafted. While MLB provides signing bonus recommendations to each team, there are no punishments if the signing bonus exceeds MLB's recommendation, unlike the NFL and NBA which have effective bonus restrictions. Also unlike the NFL and NBA drafts, there are no self-selection or commitment issues in the MLB draft. Amateur baseball players do not have to "declare" for the draft to be eligible and they often use professional agents in the bargaining process without losing their college eligibility.⁴

Thus, unlike other drafts, the MLB draft is a natural laboratory for the study of bargaining. Each draft pick initiates a simple, finite, one-shot bilateral negotiation between the team and the player it drafted. The value of the player to the team and each side's alternatives if no agreement is reached are relatively easy to model with observable information. Most importantly, the outcome of a successful negotiation (in most cases, a minor league contract and signing bonus) is publicly observable and easily quantified.

Furthermore, recent changes in the rules governing the draft provide an opportunity to study how changes in the rules of a bargaining game affect the outcome. In MLB's 2007 collective bargaining agreement (CBA) with the player's union, changes were made to the draft with the stated intent of lowering signing bonuses. These changes (the improvement of compensation to the team if a player does not sign, the reduction of MLB's bonus recommendation, and a compression of the signing period) affect the negotiations to differing degrees depending on the pick and allow one to separately measure the effect of each change.

This article presents a model of the bargaining game between the team and the player that fits the negotiation rules established by the draft process. This theoretical model is used to motivate two econometric models: A model of the probability that the player and team reach an agreement and a model of the bonus negotiated if an agreement is reached. These models are estimated using data on the top 300 draft picks in each draft from 2003-2010. The factors that the theoretical model suggests are important in reaching an agreement and determining the size of the bonus (the value of the player to the team and each side's alternatives if an agreement is not reached) are found to be significant determinants of the probability of reaching an agreement and the negotiated bonus.

The probability and bonus models are also used to estimate the effect of each change in the 2007 CBA. Consistent with bargaining theory, the compression of the signing period led to larger bonuses. Contrary to bargaining theory, the improvement

in compensation to the team if a player does not sign had little effect on the bonuses for the drafted players subject to this change. Contrary to the expectations of MLB team owners, the changes introduced in the 2007 CBA collectively resulted in larger negotiated bonuses. According to the bonus model, the rule changes increased the compensation paid to drafted players by roughly \$178 million between 2007 and 2010 (or roughly \$44.5 million per year). Despite the nonbinding nature of MLB's bonus recommendations, the bonus recommendations significantly reduced the negotiated bonuses for the players subject to them. Thus, MLB team owners would have been better off reducing the bonus recommendations without the draft changes. However, the effectiveness of the bonus recommendations has declined over time so that MLB's reduction of the bonus recommendations in 2009 and 2010 had a smaller effect on the bonuses than their identical reduction in 2007.

The article is structured as follows. The second section describes the MLB First Year Player Draft: its history, rules, and the changes introduced in the 2007 CBA. The third section describes the bargaining game used to model the draft and its implications. The fourth section describes the econometric models of the probability that an agreement will be reached and the bonus that is negotiated if an agreement is reached. The fifth section describes the data used to estimate these models and the last section describes the results.

MLB's First Year Player Draft

Throughout the history of professional baseball, team owners have attempted to restrict the compensation given to players. The most famous of these restrictions was the reserve clause that assigned a player's rights in perpetuity to the team that originally signed him. These rights could be transferred to other teams via trade or sale but never accrued to the player. Before the reserve clause was abolished, a player was only a free agent (able to shop his talents to the highest bidder) in the period before he initially signed to play professional baseball. While the reserve clause kept the compensation in annual contracts relatively low, talented amateur baseball players often received large signing bonuses for their initial contract, reflecting the bidding of multiple teams for the player. For example, in 1964, the year before the first draft, MLB teams in aggregate spent less on major league contracts than on signing amateur players.

Before the draft was instituted, team owners attempted to restrict the bonuses given to newly signed baseball players using the "bonus rule" which required that players signing contracts over a certain monetary threshold be kept on the team's 25-man major league roster. This rule and its variants did not dampen the growth of signing bonuses, but instead led teams to use their benches for inexperienced, but talented players (i.e., "bonus babies") and also promoted side payments and other behaviors meant to evade the threshold. When it became clear in the early 1960s that the bonus rule and its variants were not effectively restricting signing bonuses, MLB instituted an amateur draft giving each team the exclusive right to negotiate the

initial professional contract with each of its picks. Not surprisingly, signing bonuses plummeted with the introduction of the draft. In 1964, the year before the advent of the draft, the Los Angeles Angels signed outfielder Rick Reichardt for a record signing bonus of \$205,000. In 1965, the first pick of the first draft, Rick Monday, signed with the Kansas City Athletics for \$104,000. Reichardt's signing bonus record was not broken until 1979 when the Yankees signed Todd Demeter for \$208,000, an amount that was still much less than the 1964 record in real terms.⁵

While the draft has undergone some changes since 1965, most of its rules and structure have remained the same. Amateur baseball players from the United States (including Puerto Rico) and Canada (including international players who attend high school or college in the United States or Canada) are eligible to be drafted if they fall into one of the following categories:

- high school graduate who has not yet attended college;
- junior college student;
- junior from a 4-year college;
- senior from a 4-year college; and
- anyone who will be 21 or over within 45 days of the draft.

Players who are drafted and do not sign a contract can participate in the draft in the future if they continue to meet the eligibility requirements. Therefore, in most cases,⁶ drafted players can be placed into five groups, distinguished by their alternatives if they do not sign:

1. high school players, who can go to a junior college to re-enter the draft next year or to a 4-year college to re-enter the draft in 3 years (or 2 years if they turn 21 within 45 days of the draft);
2. junior college players, who can transfer to a 4-year college or continue to play in junior college, if they have played only 1 year at the junior college;
3. draft-eligible freshmen and sophomores (i.e., players who will be 21 within 45 days of the draft) who can continue to play in college and re-enter the draft in each subsequent year of eligibility;
4. college juniors who can re-enter the draft one more time as seniors; and
5. college seniors who, effectively, are limited to non-baseball employment if they do not sign.⁷

The only U.S. and Canadian amateur players who can sign professional contracts outside of the draft (i.e., amateur free agents) are players who were eligible to be drafted but were not.⁸ Table 1 lists the players selected in the first 300 selections of each draft between 2003 and 2010 by type.⁹

After a player is drafted, the team that selected him has exclusive negotiation rights with him for a fixed period of time (more on this below). The team is required to make the first contract offer to the player within 15 days of the draft. Subsequent

Table 1. Drafted Players by Type.

Type	2003	2004	2005	2006	2007	2008	2009	2010	Total
High school	100	91	98	92	106	92	102	112	793
Junior college	19	24	19	24	17	30	24	25	182
Freshman/sophomore	10	14	7	10	11	6	7	6	71
Junior	119	128	135	137	136	155	138	129	1,078
Senior	47	42	39	36	29	17	27	27	263

counteroffers from both sides, if necessary, can occur at any time within the negotiation window. At the end of the negotiation period, if the player has failed to sign, he re-enters the following draft if he remains eligible. The team that drafted him, but failed to sign him, cannot select him again in subsequent drafts unless the player consents in writing to be drafted by the team again. Draft picks cannot be traded and players drafted cannot be traded until a year after they are drafted. Thus, each draft pick initiates a finite, one-shot bargaining game between the player and the team that drafted him.

If an agreement is reached, it is usually in the form of a minor league contract and signing bonus. Signing bonuses can range from multiple millions of dollars for first-round picks to \$1,000 for some picks in the later rounds. The annual salaries associated with minor league contracts are capped and relatively modest compared to early-round signing bonuses. Occasionally, an early-round drafted player and team will agree to a major league contract, which necessitates that the player be kept on the team's 40-man major league roster. Major league contracts usually include a signing bonus and a guaranteed annual salary which together can sum to multiple millions of dollars. Between 2003 and 2010, 25 drafted players signed a major league contract as their first professional contract.

Beyond the exclusive negotiation rights conferred by the draft, MLB uses other means to try to keep signing bonuses to a minimum. First, for some early-round picks, MLB offers compensation to the team if it fails to sign the selected player, in the form of additional picks in the subsequent draft (more on this below). Second, MLB issues nonpublic bonus recommendations for each pick in the first five rounds of the draft. These recommendations, often referred to as "slots," are nonbinding. To exceed the slot for a pick, a team need to only notify MLB that it will exceed the slot before signing the player. MLB cannot fine or punish a team for signing a player "above slot." However, the year-to-year growth in bonuses has slowed since MLB started issuing slotting recommendations in 2002 (See Figure 1).¹⁰

The rules governing the draft are part of the CBA negotiated between MLB and the MLB Players Association (MLBPA). In the CBA negotiated in late 2006, two changes were made to the draft and subsequent contract negotiation process with the stated intent of reducing signing bonuses.¹¹ First, the draft rules were altered to give better compensation to teams who fail to sign their top draft picks. In the CBA governing the 2003-2006 drafts, if a first-round draft pick failed to sign, the team drafting him received a compensation pick at the end of the supplemental first

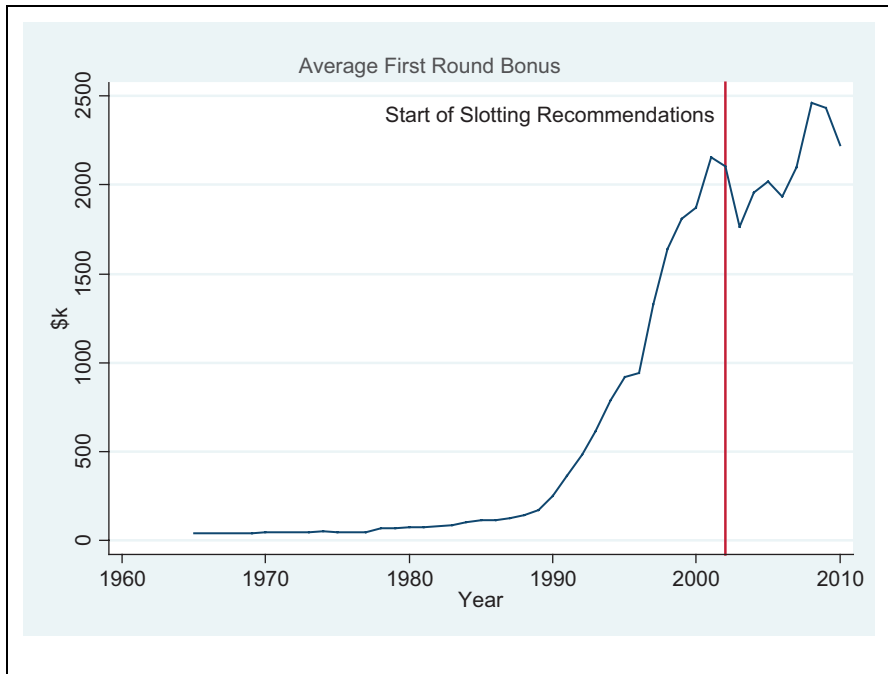


Figure 1. Average first-round bonus.

round (also called the first “sandwich” round because it is between the first and second rounds) in the following year’s draft. For example, in the 2004 draft, the Baltimore Orioles selected pitcher Wade Townsend from Rice University with its first-round selection, the eighth overall pick. The Orioles and Townsend failed to reach an agreement on a contract and, as compensation, the Orioles received the last pick in the first supplemental round (the 48th overall pick) of the 2005 draft. Teams unable to sign players selected in all rounds beyond the first received no compensation. In the CBA governing the 2007-2010 drafts (2007 CBA), if a first or second-round pick fails to sign, the selecting team receives the following pick in the next draft. For example, in the second round of the 2007 draft, with the 69th overall pick, the Atlanta Braves selected University of Georgia pitcher Joshua Fields. The Braves and Fields failed to reach an agreement on a contract, so the Braves were awarded the 70th pick of the 2008 draft. In addition, unsigned picks in the third round now result in a compensation pick at the end of the third round in the following year’s draft. In both the old and new CBA, if a player selected with a compensation pick fails to sign, the team does not receive any compensation.

The second change in the 2007 CBA pertains to the fixed time period in which the team and the drafted player must reach an agreement on a contract. In the 2003-2006 CBA, a contract could be signed at any time after the draft up until 1 week prior to

the subsequent draft. If the team and the player failed to reach an agreement 1 week before the subsequent draft, the player would go back into the player pool for the subsequent draft, if still eligible. In the 2007 CBA, the negotiation window was shortened, ending on August 15, roughly 2 months after the draft.¹² This new signing deadline effectively eliminated the “draft-and-follow” process. In the drafts prior to 2007, teams would often select a player (usually a high school player) and, if he subsequently enrolled in a junior college, follow his progress the following spring. If the player’s perceived value increased during this extra season of amateur baseball, the team would attempt to sign the player, usually for an amount above that typical for the overall selection. The draft-and-follow process was possible because teams retained the exclusive rights to a player if he enrolled in a junior college after his selection. However, if a player enrolled and attended classes at a 4-year college in the fall term after being selected, this would automatically void the drafting team’s exclusive negotiating rights, and the player would go back into the player pool for the subsequent draft, if eligible. Thus, under the 2003–2006 CBA, a high school player could unilaterally decide whether to eliminate any possibility of signing a contract with the team selecting him by attending a 4-year college or leave open the possibility of signing a contract with this team by attending a junior college. Since most fall college classes begin after August 15, the new deadline eliminated this option. It is important to note that this change does not affect college seniors who can continue to negotiate with the drafting team up to 1 week before the subsequent draft as before.¹³

While not officially part of the new CBA, a third change occurred with the 2007 draft. Possibly in anticipation of the effects of the first two changes or simply to encourage smaller bonuses, MLB instituted reduced bonus recommendations (“slots”) for the 2007 draft. While these are not public, Baseball America estimated roughly a 10% reduction in the slots between 2006 and 2007. In 2008, MLB reversed course and increased the slots back to their 2006 levels. In 2009, citing the poor economy, MLB again reduced the slots to their 2007 levels and kept them at this level in 2010.

These three changes each affect a different subset of the draft class in the drafts conducted under the new CBA. The improvement in compensation if a player does not sign only applies to players selected in the first three rounds (including the first supplemental round), except for compensation picks for players who failed to sign the previous year. The contraction of the signing period only affects high school, community college, and college underclass players, not college seniors. The slots only apply to picks in the first five rounds (including compensation picks). Because the changes affect different players, it is possible to estimate the effect of each change on the negotiated bonuses.

Bargaining Model of the First Year Player Draft

The draft rules described in the previous section can be modeled with a relatively simple bargaining game. The draft gives each team the exclusive, nontradable right

to negotiate with each player it drafts, so the negotiation process triggered by each draft pick is a simple bilateral negotiation. Each negotiation has a deadline and, if no agreement is reached, the team cannot select the player again in subsequent drafts, so the bargaining game between the team and drafted player is a finite, one-shot game.

Let v_{ij} be team j 's valuation of player i . This valuation can be thought of as the team's expectation of the net present value of the player's cumulative marginal revenue product. This valuation is assumed to be known by both the player and the team, as it is based on publicly observable characteristics of the player. Any special value assigned to a player by a team is largely revealed through the team's act of drafting the player. $\bar{v}_{ij}$ is the "no agreement" value of player i for team j . In other words, $\bar{v}_{ij}$ is the value of the compensation that team j receives if it does not sign player i . This value is also assumed to be common knowledge and largely a function of the perceived strength and depth of next year's draft class if compensation exists for the player; otherwise, it equals zero.

Let w_i be player i 's "no agreement" value (i.e., the value of the next-best option for the player if the player and the team fail to sign a contract). This value is assumed to have a common knowledge component that is a function of the player's characteristics (X_i) and also a component (ϵ) that is private knowledge to the player (e.g., a high school player's perceived utility of the "college experience"), but effectively stochastic from the team's (and the econometrician's) point of view: $w_i = g(X_i) + \epsilon$. Because the draft has no commitment devices (e.g., "declaring" for the draft as in the NFL and NBA), the player has no credible way to reveal ϵ before the draft.

Under MLB's post-draft negotiation rules, the team is required to make the first offer within 15 days of the draft. Because the team cannot use strategic delay, the player has an advantage in the negotiation. If only one offer could be made in each discrete time period, the unique equilibrium would closely resemble the famous "ultimatum game." After the team makes its initial offer, the player has no incentive to make a counteroffer until just before the deadline. If $v_{ij} - \bar{v}_{ij} > w_i$ (so there exists a post-draft net benefit to be split between player i and team j) and both sides are risk neutral, the unique solution to this negotiation would involve player i making a counteroffer of an amount slightly less than $v_{ij} - \bar{v}_{ij}$ in the last period before the negotiation deadline and the team accepting this offer. If the player makes a counteroffer far enough before the deadline to give the team an opportunity to make a counter-counteroffer, the team will wait until the period just before the deadline and offer the player a bonus just slightly more than its expectation of w_i . Therefore, if the player rejects the team's initial offer, the player will wait until the period just before the deadline and offer a bonus slightly less than $v_{ij} - \bar{v}_{ij}$ and the team will accept. Given this, the team will make an initial offer of (slightly less than) $\delta(t) (v_{ij} - \bar{v}_{ij})$ and the player will accept, where $\delta(t)$ is the player's discount factor as a function of t , the total time of the negotiation period.

The preceding is the unique subgame perfect equilibrium if only one offer and acceptance/rejection can be made in each discrete time period. In an actual post-draft negotiation, offers and counteroffers may be made at any frequency. Ma and Manove (1993) show that a similar equilibrium obtains in negotiations with a deadline if offers can be made at any frequency and there is a random delay between offers and the acceptance/rejection decision. Such a delay could come from a delay in receiving the offer (e.g., not being available by phone when the offer is made) or from the time it takes to consider an offer. Ma and Manove (1993) show that negotiating parties will wait to make an offer until a point in time when there is a sufficient probability that the random delay will prevent a decision before the deadline. Adapting their model to the MLB draft negotiations, a player will delay his counteroffer until close to the deadline to minimize the chance that the team will be able to optimally counter-counteroffer, but not so close to the deadline that the team likely would be unable to accept the offer after a random transmission or decision delay. Unlike the ultimatum game above, in the Ma and Manove (1993) bargaining game, agreements are not reached until just before the deadline and there is a small positive probability that no agreement is reached even when there is a positive surplus because the player's timing of the offer plus the random delay may exceed the deadline.

This quasi-ultimatum aspect of the negotiations between the team and the drafted player is well recognized in the industry. Many agents wait until late on the day of the negotiation deadline to make the initial counteroffer. As one general manager stated, "deals got done at 11:59 [on August 15] with literally no conversations for long periods of time."¹⁴ From an agent's point of view, "If I'm advising a first-round pick, I'm telling him there's no reason to sign before August 15. The longer you wait, the more money you'll get."¹⁵

As shown by Ma and Manove (1993), if both participants are risk neutral and of equal ability, the one making the final offer will receive the majority of the surplus available. However, teams and players may be risk averse and the negotiating abilities of the players, agents, and teams may vary. Therefore, the equilibrium is characterized as a generalized bargaining equilibrium in which the bonus is a weighted average of $v_{ij} - \bar{v}_{ij}$ and w_i : $b_{ij}(w_i, v_{ij}, \bar{v}_{ij}) = \alpha_{ij}(v_{ij} - \bar{v}_{ij}) + (1 - \alpha_{ij})w_i$ if $v_{ij} - \bar{v}_{ij} > w_i$, where $\alpha_{ij} \in (0, 1)$ reflects player i 's relative advantage in the negotiation with team j .¹⁶

The equilibrium above is confined to the post-draft negotiation between the player and the team that drafted him. However, this stage game is played in the context of a larger meta-game in which an amateur player might be drafted multiple times. Using backward induction, consider first the negotiation that occurs when a senior is drafted. In almost all cases, $w_i = 0$ for a senior, as his only outside option is non-baseball employment. Therefore, for a senior drafted player, the negotiated bonus is $b_{ij}^{sr} = b_{ij}(0, v_{ij}, \bar{v}_{ij})$. For a drafted college junior, the next best outside option if an agreement is not reached is returning to his final year of college

eligibility and re-entering the draft next year as a senior. Re-entering next year's draft entails significant risk as the player's perceived value may increase or decline in his senior year. The value of a college junior's outside option is:

$$w_i^{jr} = \delta(1) E\left(b_{ij}\left(0, E\left(v_{ij}^{d+1}\right), E\left(\bar{v}_{ij}^{d+1}\right)\right)\right) + \epsilon \quad (1)$$

where $E\left(v_{ij}^{d+1}\right)$ is the player's expectation of his perceived value in next year's draft, $E\left(\bar{v}_{ij}^{d+1}\right)$ is the player's expectation of the value of the compensation available to the team drafting him next year, and $E\left(b_{ij}\right)$ is the player's expectation of the bonus he can negotiate next year evaluated across all of the teams that might draft him next year. While w_i^{jr} might be zero or negative for some players (e.g., those who dislike college), on average $w_i^{jr} > 0$ as the player's expectation of his value next year should be the same as his value this year. Likewise, the value of a junior college or sophomore player's outside option can be characterized similarly and will, on average, be greater than a junior's outside option value. Similarly, a high school player's outside option value will be greater, on average, than all others. In other words, a player's outside option value will be greater the farther he is from the point at which he has no baseball-related outside options as a drafted college senior. Thus, all else equal, $b^{hs} > b^{jc/so} > b^r > b^{sr}$. Furthermore, college seniors should reach agreement on a contract with almost perfect certainty as $v_{ij} - \bar{v}_{ij} > 0$ (otherwise the player would not be drafted), whereas high school players are least likely to sign as they are the players for which $v_{ij} - \bar{v}_{ij} < w_i$ is most likely.

Although simple, this model offers many testable implications concerning the negotiated bonuses. First, the bonus increases with the value the player brings to the team. Second, the bonus decreases with the value of the team's alternative if no agreement is reached. This implies that bonuses should fall as the compensation given to the team (if the player does not sign) improves. Third, the bonus increases with the player's outside option. This implies that, all else equal, high school players will negotiate the highest bonuses and college seniors will receive the lowest bonuses, as summarized above. Finally, the model implies that the compression of the signing period will lead to larger bonuses as a player can implement the "ultimatum" strategy sooner. As described above, the present value of the player's bonus is bounded above by $\delta_i(t) (v_{ij} - \bar{v}_{ij})$, which is a decreasing function of t , the length of the signing period.

Because the slots are nonbinding, the model implies that they will have no effect on the signing bonuses. However, some behavioral economics research suggests that reference or focal points, like the slots, impart significant "framing" effects on the outcomes of economic activity.¹⁷ Despite their nonbinding nature, the slots (where present) may serve as a benchmark for each negotiation and the increase or decrease of these recommendations each year could have significant effects on the negotiated bonuses. The noticeable reduction in the growth of the

average first-round bonus after the slots were introduced in 2002 seems to support this argument.

The model described above also has many testable implications concerning the probability that a drafted player and team sign a contract. The player and team will fail to sign a contract when (1) the player's offer plus the delay in communicating/processing the offer exceed the deadline, or (2) $v_{ij} - \bar{v}_{ij} < w_i$. This implies that seniors are most likely to sign a contract, while players with better outside options, such as high school players, are less likely to sign. In addition, an improvement in the value of the team's outside option, $\bar{v}_{ij}$, will reduce the probability that a contract will be signed. Equation 1 and the analogous conditions for high school players and junior college players imply that first round picks are more likely to sign than later picks. For example, if a player is drafted in the first round, his perceived value is unlikely to improve in the next draft if he fails to sign. However, a (nonsenior) player drafted in a lower round may decline a contract and gamble that his future value will be higher in the subsequent draft.

Empirical Model

As discussed above, the negotiated bonus can be modeled as a linear function of the value of the player to the team and the alternatives of each side if no agreement is reached. The value of a player to a team is the present discounted value of his expected marginal revenue product for that team. A player may increase a team's revenue in a number of ways, but a player's primary revenue contribution to a team is by helping it win more games. Many studies have shown that demand increases as a team's winning probability increases.¹⁸ Jazayerli (2005) and Andrecheck (2009) both found that there is a declining and nonlinear relationship between a player's overall draft position and his expected aggregate wins above replacement (WAR).¹⁹ Therefore, the value of a player to a team is assumed to be a nonlinear (specifically, a fifth-order polynomial) function of the player's overall pick:

$$v_{ij} = \sum_{k=1}^5 PICK_i^k + \phi_j \quad (2)$$

where the last term, ϕ_j , reflects differences in the value of drafted players across teams (e.g., due to differences in the marginal revenue of a win or differences in the perceived value of amateur players). In alternate specifications of the bonus model, these team effects are replaced with the team's franchise value ($VALUE_j$) or the team's total revenue ($REVENUE_j$) during the year contemporaneous with the draft.

Because the value of a player may differ across teams and draft picks cannot be traded, players are not always drafted in declining order of their overall talent. For example, a small-market team, for which the marginal revenue of a win is relatively low, may estimate that $v_{ij} - \bar{v}_{ij} < w_i$ for the most talented player available at its draft

position. This team may instead select a less-heralded player, but one with a lower w_i , which it can sign for a smaller bonus. On the other hand, teams from large markets, for which the marginal revenue of a win is relatively high, may select high talent, high w_i players that “fall” to them in the draft and sign them for a larger bonus than consistent with the draft pick. To capture this phenomenon, the constructed variable denoted as $SLIDE_i$:

$$SLIDE_i = \frac{PICK_i - RANK_i}{RANK_i} \quad (3)$$

is included as an independent variable, where $RANK_i$ is player i 's pre-draft ranking according to Baseball America. Players with a positive value for $SLIDE$ are those who fell below their pre-draft rank before being selected. Players with a negative value for $SLIDE$ are those selected above their pre-draft rank.

The outside options of players depend on their type (e.g., high school, community college, sophomore, junior, or senior), so dummy variables for the player's type (HS , $CCSO$, and SR) are also included in the bonus model. Community college players are grouped with draft eligible freshmen and sophomores due to the small number of freshmen and sophomores and because both groups may re-enter the draft multiple times if they do not reach an agreement with the drafting team. Player-type effects are measured relative to college juniors.

The outside options of the teams are the draft picks received in the subsequent year's draft if they do not sign the current drafted player. The value of these compensation picks is measured with a series of dummy variables. The $COMP1$ dummy variable equals one for first-round selections with compensation. $COMP1_NEW$ equals one for first-round selections with compensation in years 2007-2010. Its estimated coefficient will measure the effect under the new CBA of improving the compensation for first-round picks who do not sign from “end of supplemental round” compensation to “pick+1” compensation. $COMP2$ equals one for supplemental and second-round picks with compensation and $COMP3$ equals one for third-round picks with compensation. Recall that the new CBA added “pick+1” compensation in the subsequent draft for supplemental and second-round picks who do not sign and “end of round” compensation for third-round picks who do not sign. The estimated coefficients for the latter two variables will measure the effects of these changes.

To capture the effect of the compressed signing period in the new CBA, the dummy variable $NEW_DEADLINE$ is included in the model. This variable is equal to 1 for all nonseniors in 2007-2010 and 0 otherwise. To investigate whether the effect of the new deadline is greater in the earlier rounds, a second model includes $NEW_DEADLINE$ interacted with the first round ($NEW_DEADLINE1$), interacted with the supplemental and second rounds ($NEW_DEADLINE2$), and interacted with all other rounds ($NEW_DEADLINE_ABOVE$).

To capture the effect of the slots and any year-to-year changes in these recommendations on the negotiated bonuses, a dummy variable for picks subject to slotting

Table 2. Continuous Variables.²⁵

Variable	N	Mean	SD	Min	Max
BONUS (\$k)	2260	494.7	743.5	1	7500
SLIDE	2400	-0.02	0.80	-0.93	11
VALUE (\$m)	240	425.4	220.4	145	1700
REVENUE (\$m)	240	172.2	49.4	80	441

(i.e., those in the first five rounds, including compensation picks) interacted with the dummy variable for each year (*SLOTPICK * YEAR*) is included. Also included in the bonus model as covariates are dummies for major league contracts (*ML*), for draft-and-follow selections (*DF*), and for each year to capture any differences in the talent and depth of each draft pool.

The dependent variable in the bonus model is the signing bonus (*BONUS*), if the player signed a minor or major league contract.²⁰ The bonus model is estimated, conditional on signing a nonvoided contract, using fixed effects ordinary least squares (OLS) with clustered robust standard errors, where the clusters are the teams, reflecting the possibility that bonus negotiations between a team and its draft picks may not be independent.

The model of the probability of signing a contract is a probit model that incorporates the factors which may lead to a breakdown in negotiations. As in the bonus model, type dummies (*HS*, *CCSO*, and *SR*) are included. *PICK* is included linearly to test the hypothesis that the probability of signing a contract decreases for picks in later rounds. *SLIDE* is included to test the hypothesis that players who fall in the draft below their pre-draft ranking are less likely to sign. To test whether the changes in the new CBA changed the probability that a player and team would reach an agreement, *COMP1*, *COMP1_NEW*, *COMP2*, *COMP3*, and *NEW_DEADLINE* are also included in the probability model. The dependent variable of the probability model is the dummy variable *SIGNED*. Tables 2 and 3 list the summary statistics for variables found in either the bonus or probability model, apart from *PICK* and *RANK*.

Data

The primary source of data for this analysis is Baseball America's Advanced Draft Database (ADD) for each draft from 2003-2010.²¹ The ADD contains the following information for each player drafted: the player's name, the player's school, the player's position, the draft round, the selection number (i.e., the overall pick), the team making the selection, whether the player signed a professional contract, and, if so, the signing bonus associated with the contract. In most cases, drafted players are signed to minor league contracts which include a signing bonus and a relatively small annual salary.²² Between 2003 and 2010, 25 drafted players signed a major

Table 3. Dummy Variables.

Variable	Mean	Variable	Mean
SIGNED	0.943	SR (senior)	0.110
COMPI	0.100	SLOT PICK2003	0.065
COMPI_NEW	0.050	SLOT PICK2004	0.067
COMP2	0.085	SLOT PICK2005	0.071
COMP3	0.050	SLOT PICK2006	0.069
NEW_DEADLINE	0.458	SLOT PICK2007	0.077
NEW_DEADLINE1	0.050	SLOT PICK2008	0.072
NEW_DEADLINE2	0.085	SLOT PICK2009	0.071
NEW_DEADLINE_ABOVE	0.322	SLOT PICK2010	0.073
HS (high school)	0.330	DF (draft and follow)	0.002
CCSO (community college-sophomore)	0.105	ML (major league contract)	0.010

league contract as their first professional contract. Major league contracts require that the player be placed on the team's 40-man major league roster and usually include a signing bonus, a significant guaranteed salary, and a fixed term. For these 25 players, the total guaranteed amount (i.e., bonus plus salary over the life of the contract) is taken from Baseball America's 2009 Almanac and other media reports.

The data are restricted to the top 300 picks in each draft. This roughly corresponds with the first 9 or 10 rounds of the draft (depending on the length of the supplemental rounds between the first four rounds). Beyond the tenth round, the players selected are usually marginal prospects and the bonuses paid to sign them are usually small with little variance across players. As described above, an exception to this rule, at least before 2007, were draft-and-follow players: players who were lightly regarded at the time of the draft, but improved their perceived worth during the subsequent year. Before 2007, these players, who were typically drafted in rounds 11–50, could sign with the team that drafted them (up to a week before the subsequent draft) or re-enter the draft. As players with improved perceived value, draft-and-follows received bonuses much higher than typical for their round. There are 4 draft-and-follow players in the roughly 2400 players used for this analysis.

The ADD data are supplemented with Baseball America's pre-draft player rankings published the week before the draft, reflecting the scouting community's general consensus about each player's relative worth. Baseball America's (BA's) pre-draft rankings only include the top 200 draft-eligible players. The first nonranked player selected in each year is assigned a rank of 201, the second a rank of 202, and so on.

For college players, the ADD data are supplemented with information on the player's class (e.g., junior college, freshman, sophomore, junior, or senior). This represents the player's college baseball eligibility, which is not necessarily the same as their academic class. For example, a sophomore has 2 years of college eligibility left and could re-enter the draft up to two more times if he does not sign. A junior has 1 year of eligibility left, and a senior has exhausted his college baseball

eligibility. The information on the player's class is taken from various sources: BA's pre-draft rankings (for ranked players), BA's pre-draft scouting reports, the Baseball America Almanac for the draft's year (which lists the class and statistics for some college players), The Baseball Cube (which lists college statistics for some drafted players),²³ mlb.com, and the baseball pages on various college websites. Class information was unavailable for six players (five from 2003 and one from 2004).

While MLB's bonus slots are not public, the slots for 2006-2010 as estimated by Jim Callis at Baseball America are also included.²⁴ The estimate of each team's franchise value and revenues for each year from 2003-2010 is taken from Forbes.com.

Results

Table 4 lists the estimated coefficients and standard errors for the bonus model. The first column lists the coefficient estimates and clustered standard errors for the primary model and the second column lists the coefficient estimates when the new deadline dummy is interacted with the first round, second round, and all other rounds. (Team fixed effects are not reported in the table.) Pooled OLS estimates in which the team effects are replaced by the team's franchise value and revenue, respectively, are reported in Appendix A.

As discussed above, previous research has shown that a drafted player's expected value declines nonlinearly with his pick. The estimated coefficients of the *PICK* polynomial confirm that signing bonuses follow the same pattern. The bonus model also confirms that high school players negotiate larger bonuses, reflecting their superior bargaining position. All else equal, high school players receive bonuses that are roughly \$110,000 higher on average than college juniors. Seniors, who have few baseball-related options if they do not sign, receive the smallest bonuses of any type of drafted player.²⁶

Players who "slide" in the draft command higher bonuses as well, since their selection underestimates their talent and they are more likely to not sign a contract. Many baseball analysts have argued that this aspect of the draft subverts its "competitive balance" function. Small market teams, whose marginal revenue from a win is relatively small, may estimate that the net benefit they will receive from a top player is less than the expected value of the player's options if he does not sign. They may bypass such players, letting them fall to large market teams who sign them with bonuses larger than consistent with the overall selection. In this way, the draft may reinforce the divide between winning and losing teams, instead of ameliorating it. However, the correlation between *SLIDE* and the team franchise value (*VALUE*) and revenue (*REVENUE*) is close to 0 (.06 for both). Teams that aggressively draft and sign highly ranked players that fall to them are as likely to be large market teams as small market teams. Especially in recent years, the teams with the highest mean *SLIDE* values include small market, losing teams like Kansas City and Pittsburgh as well as large market, winning teams like Boston.

Table 4. Bonus Model.²⁷

Model	(1)	(2)	Model	(1)	(2)
PICK	-130.16*** (10.99)	-129.64*** (10.89)	NEW_DEADLINE_ABOVE		85.08** (31.62)
PICK ²	1.87*** (0.18)	1.87*** (0.18)	SLOT PICK_2003	-184.82*** (39.60)	-179.32*** (37.39)
PICK ³	-0.013*** (0.001)	-0.013*** (0.001)	SLOT PICK_2004	-140.45*** (25.92)	-134.98*** (23.35)
PICK ⁴	4.0×10^{-5} *** (4.3×10^{-6})	3.9×10^{-5} *** (4.3×10^{-6})	SLOT PICK_2005	-146.31*** (28.05)	-141.06*** (26.61)
PICK ⁵	-4.6×10^{-8} *** (5.2×10^{-9})	-4.6×10^{-8} *** (5.1×10^{-9})	SLOT PICK_2006	-173.18*** (37.68)	-167.91*** (37.89)
SLIDE	149.36*** (14.40)	149.89*** (14.42)	SLOT PICK_2007	-315.51*** (52.53)	-314.57*** (53.02)
ML (major league contract)	832.47*** (298.98)	837.75*** (292.82)	SLOT PICK_2008	-163.58*** (52.94)	-163.28*** (52.91)
DF (draft and follow)	209.96 (228.49)	207.57 (224.71)	SLOT PICK_2009	-183.02*** (49.03)	-185.54*** (48.06)
HS (high school)	114.82*** (19.10)	113.72*** (19.11)	SLOT PICK_2010	-165.60*** (54.23)	-168.30*** (54.37)
CCSO (community college- sophomore)	58.57*** (16.65)	58.39*** (16.60)	YEAR = 2004	-12.71 (11.17)	-12.87 (11.24)
SR (senior)	-16.55 (19.12)	-19.04 (17.77)	YEAR = 2005	19.49* (10.98)	19.30* (11.11)
COMPI	-483.55*** (117.75)	-470.82*** (118.18)	YEAR = 2006	18.60* (9.42)	18.34* (9.71)
COMPI_NEW	387.26*** (126.49)	236.96 (306.07)	YEAR = 2007	-29.65 (36.26)	-17.56 (26.32)
COMP2	91.95*** (41.74)	43.23 (105.81)	YEAR = 2008	-20.94 (38.31)	-7.74 (30.05)
COMP3	28.73 (28.27)	35.99 (24.74)	YEAR = 2009	-12.95 (36.84)	-0.42 (29.13)
NEW_DEADLINE	100.08*** (40.02)		YEAR = 2010	4.49 (36.20)	16.24 (28.22)
NEW_DEADLINEI		250.62 (283.80)	Observations	2,260	2,260
NEW_DEADLINEZ		147.07 (105.36)	R ²	.83	.83

Table 5. Average Residual/Predicted and Premium Over Slot for Nonseniors³¹ in 2009-10, Rounds 1-5.

Time of Signing	Number	Number in First Round	Average Residual/ Predicted	Average Premium Over Slot
Friday or before	226	29	-13%	9%
Weekend to 5 pm on deadline day	32	7	30%	173%
After 5 pm on deadline day	48	22	43%	130%

As theory predicts, the compensation given to teams if they fail to sign a drafted player (COMP1) reduces the negotiated bonus significantly, as the team's negotiating position is better with this compensation. However, the coefficient estimate on COMP1_NEW implies that the improvement in the first-round compensation with the new CBA *increased* the negotiated bonuses. Furthermore, the introduction of compensation where none existed before (COMP2 and COMP3) also increased bonuses, although the latter coefficient is not statistically significant. These latter findings contradict bargaining theory. One would expect (as did MLB owners) that improving the compensation given to teams and introducing compensation where none existed before for players who do not sign would lead to smaller bonuses due to the team's improved negotiating position.

One possibility is that the coefficient for COMP1_NEW is picking up some of the effect of the new deadline, particularly if the effect of the new deadline is larger in the first round. As seen in Table 5, there is some evidence that players in the first round are more likely to implement the ultimatum strategy than players in lower rounds, so the effect of the new deadline may be greater for first-round players. To test this, the second model includes interaction terms between the new deadline dummy and the first round (NEW_DEADLINE1), the second round (NEW_DEADLINE2), and all subsequent rounds (NEW_DEADLINE_ABOVE). The inclusion of the interaction terms reduces the estimated COMP1_NEW coefficient and it is no longer statistically significant. This is likely due to the fact that COMP1_NEW and NEW_DEADLINE1 are highly collinear (correlation coefficient = .96).²⁸ Thus, there may be insufficient information in the data to fully separate the effect of the new deadline from the effect of the improvement in compensation, especially for early-round picks.

If the primary cause of the larger bonuses for early-round picks subject to the new deadline is the increased use of the ultimatum strategy, the estimates in Table 4 may indicate that what is important is the existence of compensating draft picks in the subsequent draft, not the level of the compensating picks. One hypothesis is that teams do not view "pick+1" compensation as much of an improvement over supplemental round compensation for first-round picks because pick+1 compensation in the first round would most likely lead to multiple expensive first-round selections

in the subsequent draft if the current first-round drafted player does not sign. In other words, the team may feel forced to “underdraft” with its compensation draft pick for budget reasons, leading the team to view “pick+1” compensation as no different than supplemental round compensation. For example, the Washington Nationals failed to reach an agreement with their first-round pick (#9 overall) in 2008, Aaron Crow, and received the 10th pick in the first round of the 2009 draft as compensation. They used this pick to select Drew Storen, who was not viewed by most as a first-round talent at the time,²⁹ and signed him a few days after the draft with a bonus \$263,000 less than the slot for the 10th pick. Many at the time speculated that Washington selected Storen with the 10th pick, so they could redirect their draft budget toward signing their other first-round pick (#1 overall), Stephen Strasburg, who eventually signed with the Nationals under a major league contract with a bonus \$3.5 million over the slot for the first overall pick. In any event, underdrafting for budget reasons would not explain why the introduction of compensation in the supplemental, second, and third rounds led to larger bonuses.

As measured by the NEW_DEADLINE coefficient, the compression of the signing period increased the negotiated bonuses, as bargaining theory predicts. The new deadline leads to larger bonuses because it makes it less costly for players and their agents to employ an ultimatum strategy (i.e., waiting until just before the deadline to make a counteroffer) which benefits the players. For the 2009 and 2010 drafts, I tracked when the team and player reached an agreement on a contract using media reports from Baseball America and other sources. Table 5 lists the residual as a percentage of the model’s predicted value,³⁰ denoted as Residual/Predicted, and the percentage by which the bonus exceeded the slot ((BONUS – SLOT)/SLOT), denoted as “Premium Over Slot,” for players drafted in the first five rounds who were subject to the new deadline and signed a contract.

As seen in Table 5, those players who waited to sign until just a few days before the deadline received higher bonuses on average than those who signed earlier. Those who waited until just before the deadline received bonuses 43% higher on average than the model predicts. Of course, all nonseniors benefit from the new deadline, regardless of when they sign, because the ultimatum strategy is a more credible threat than before. However, Table 5 illustrates that the ultimatum strategy is effective for those who employ it. In addition, the ultimatum strategy is used disproportionately by first-round draft picks. This may be due to the greater use of agents by first-round draft picks and the greater expertise of first-round agents. For instance, in 2009, all of the first-round draft picks were advised by professional agents, whereas only 76% of non-first-round draft picks (in the top 300 picks) were professionally advised. In addition, the Boras Corporation represented 6 of the 32 first-round picks but only 3 of the 268 non-first-round picks.

The net result of the changes from the new CBA is significantly more bonus compensation to drafted players than before.³² After controlling for the changes in the composition of the draft classes between 2003-2006 and 2007-2010 through the primary bonus model described above, the total increase in bonuses due to the changes

Table 6. Unsigned Players by Year and Round.³⁷

Round	2003	2004	2005	2006	2007	2008	2009	2010	Total Unsigned	Total Signed
1	0	1	0	0	0	2	2	3	8	236
2	0	0	1	1	2	2	3	0	9	374
3	3	0	2	2	4	2	3	1	17	231
4	1	1	1	0	4	1	0	4	12	228
5	2	1	0	1	3	2	2	2	13	227
6	3	0	3	0	1	0	3	5	15	224
7	0	4	0	3	1	2	1	4	15	225
8	2	3	5	0	4	0	2	3	19	220
9	1	1	3	5	3	2	1	2	18	218
10	2	2	0	2	0	4	0	1	11	77
Total	14	13	15	14	22	17	17	25	137	2,260

Table 7. Marginal Change in the Probability of Signing a Contract.³⁸

DV=	SIGNED
PICK	-0.00030*** (0.00006)
SLIDE	-0.01528*** (0.00352)
HS (high school)	-0.05864*** (0.01188)
CCSO (community college-sophomore)	-0.01490 (0.01583)
SR (senior)	0.03813*** (0.00814)
COMPI	0.02135 (0.02088)
COMPI_NEW	-0.10832* (0.09209)
COMP2	-0.00350 (0.01954)
COMP3	-0.03435 (0.02857)
NEW_DEADLINE	-0.00743 (.00886)
Observations	2,397
Pr (mean X)	0.961
Pseudo R ²	0.11

in the new CBA is \$178 million or roughly \$44.5 million per year. In other words, the increase in bonus compensation to drafted players was roughly \$1.5 million per team per year because of the changes made in the new CBA. This was not the expectation at the time, as the changes were a concession to the owners by the players to partly compensate the owners for expected increases in free agent compensation from other changes in the CBA. As Jim Callis summarized in a recent article:

MLB never saw this coming. When bonus spending reached \$156 million in 2006, owners sought and received adjustments to the draft in a new collective bargaining agreement. The two biggest changes were the establishment of a signing deadline and drastically improved compensation for unsigned picks in the top three rounds. MLB officials publicly said the new rules would allow teams to focus

on ability more than signability, and privately crowed that they'd give clubs extra leverage.³³

Tracy Ringolsby goes on to describe the results of this effort in another recent article:

The thought was that by not allowing negotiations to drag out for months, teams would have more leverage, and with better compensation for not signing a player, teams would be more willing to walk away. Overall, though, the signing deadline has created more of a feeding frenzy, with teams cranking up their offers at the last minute to sign their players. And the compensation has not been powerful enough, by and large, for teams to let premium talent get away. Total draft expenditures have soared. In 2010, teams handed out a combined \$194.8 million in bonuses, breaking the record for a third straight year.³⁴

Unlike the improved compensation and new signing deadline, the slotting recommendations significantly reduced bonuses according to the bonus model's estimates. As seen in Table 4, the *SLOTPICK*YEAR* coefficient is large and statistically significant for all years. Despite the nonbinding nature of the slotting recommendations, they seem to act as an effective benchmark to the negotiations. In particular, when MLB reduced the slotting recommendations by 10% in 2007, this significantly reduced the negotiated bonuses in 2007, almost doubling the negative *SLOTPICK* coefficient for 2007.³⁵ However, the effectiveness of the slots is limited. In 2008, MLB increased the slots back to their 2006 levels and then, in 2009 and 2010, reduced the slots to the lower 2007 levels. The reduction of the slotting recommendations the second time in 2009 was not as effective as before and only led to a small reduction in the bonuses.³⁶ The effect of the slots eroded further in 2010, to the point that the slots' effect on bonuses in 2010 was essentially the same as in 2008. This is likely because the slotting recommendations have become less realistic over time as they have diverged from the expected value of the picks by not keeping pace with the growth of MLB. The 2010 slots were 10% less than the 2006 slots despite the fact that MLB revenues in 2010 were 20% greater than in 2006.

Table 6 lists the number of drafted players who did not sign by year and round. While the vast majority of players drafted in the first 300 selections agree to a contract, a small number do not sign, particularly in rounds 3 through 10. The number of unsigned players also increased slightly after 2006.

Table 7 lists the estimated marginal effects (i.e., the change in the probability of signing a contract with a change in the independent variable)³⁹ from the probit model of the probability of signing a contract.

The estimated results from the probit model largely confirm the predictions of bargaining theory. Although most drafted players (in the top 300 selections) sign, the probability of signing a professional contract declines with the overall pick. As expected, high school players are less likely to sign than college juniors (who are the benchmark type in the probability model) as they have more options if they do

not sign. College seniors are the most likely player type to sign as they have few baseball-related outside options. In fact, only two college seniors drafted in the top 300 picks between 2003 and 2010 failed to sign. One of them, Jerrad Page, was also selected in the NFL draft and chose to play football professionally instead of baseball. In addition, players who “fall” in the draft (i.e., are drafted lower than their rank) are less likely to sign and those selected above their draft rank are more likely to sign. For example, a talented high school player, with a scholarship to a top college program, who falls below the draft position he expected, may reason that he is better off going to college with the hope that his draft position will be significantly better in 3 years. Although there is risk in not signing, there are many examples of high school players who have succeeded with this strategy. For example, Brian Matusz, who was ranked as the 66th best player in the 2005 draft, was drafted by the Angels in the 4th round with the 133rd overall selection. Matusz did not sign with the Angels and instead attended the University of San Diego. In 2008, he was drafted by the Baltimore Orioles with the fourth overall selection and signed a major league contract with a \$3.2 million bonus.

The probability model also provides evidence that some of the changes in the new CBA reduced the probability of signing. Consistent with the bargaining model, the improved compensation given to teams for first-round picks who do not sign and the new compensation given to teams for second- and third-round picks, who do not sign reduced the probability of signing, although only the first-round effect was statistically significant. It does not appear that the new signing deadline affected the probability of signing.

Conclusion

The implications of bargaining theory are largely confirmed by the bonuses negotiated between MLB teams and drafted players between 2003 and 2010. Bonuses declined nonlinearly with the overall selection and players with more outside options (e.g., high school students) negotiated larger bonuses than players with few outside options (e.g., seniors).

However, the evidence is mixed regarding the changes instituted with the 2007 CBA. As theory predicts, the shortening of the signing period led to larger bonuses, as it made the ultimatum strategy a more credible threat for the players, and many players, particularly in the first round, used this strategy to increase their compensation. However, the improvement of the compensation available to teams for players who do not sign a contract increased the negotiated bonuses, contrary to the predictions of bargaining theory. Overall, these two changes led to larger negotiated bonuses for drafted players, of roughly \$1.5 million per team per year. While MLB's slots acted as an effective benchmark for negotiations—significantly lowering the negotiated bonuses—their effectiveness wore off over time, after two 10% reductions in the recommendations in 2007 and 2009.

Appendix A

Table A.I. Bonus Model With Alternate Specifications.

Model	DV = Bonus or Guaranteed Compensation if Major League Contract			Pooled OLS (Value) (2003-09)	Pooled OLS (Revenue) (2003-09)
PICK	-134.98*** (13.77)	-134.35*** (13.76)	-131.60*** (10.93)	-131.62*** (10.92)	
PICK ²	1.96*** (0.23)	1.96*** (0.23)	1.90*** (0.18)	1.90*** (0.18)	
PICK ³	-0.013*** (0.002)	-0.013*** (0.002)	-0.013*** (0.001)	-0.013*** (0.001)	
PICK ⁴	4.2×10^{-5} *** (5.6×10^{-6})	4.2×10^{-5} *** (5.6×10^{-6})	4.0×10^{-5} *** (4.3×10^{-6})	4.0×10^{-5} *** (4.3×10^{-6})	
PICK ⁵	-4.9×10^{-8} *** (6.7×10^{-9})	-4.9×10^{-8} *** (6.7×10^{-9})	-4.7×10^{-8} *** (5.1×10^{-9})	-4.7×10^{-8} *** (5.1×10^{-9})	
SLIDE	149.24*** (19.30)	149.78*** (19.31)	154.53*** (14.43)	154.88*** (14.33)	
ML (major league contract)	2688.26*** (552.65)	2694.73*** (547.47)	847.90*** (297.62)	847.22*** (297.13)	
DF (draft and follow)	70.68 (311.73)	67.82 (307.73)	184.03 (240.37)	182.25 (240.97)	
HS (high school)	115.93*** (19.47)	114.73*** (19.36)	111.79*** (18.45)	112.16*** (18.53)	
CCSO (community college- sophomore)	62.82*** (18.96)	62.56*** (18.85)	52.93*** (15.25)	52.66*** (15.29)	
SR (senior)	-12.13 (22.92)	-14.92 (20.93)	-8.87 (18.71)	-8.95 (18.71)	
COMPI	-530.28*** (147.07)	-515.25*** (149.57)	-499.81*** (118.29)	-499.13*** (118.44)	
COMPI_NEW	438.14*** (158.60)	260.02 (365.79)	380.22*** (125.56)	379.63*** (125.56)	

(continued)

Appendix A

Table A1 (continued)

Model	DV = Bonus or Guaranteed Compensation if Major League Contract	Pooled OLS (Value) (2003-09)	Pooled OLS (Revenue) (2003-09)
COMP2	90.88** (43.08)	84.23* (42.28)	84.23* (42.34)
COMP3	25.73 (27.09)	23.56 (28.43)	24.17 (28.21)
NEW_DEADLINE	111.42** (48.48)	108.19*** (38.32)	108.28*** (38.31)
NEW_DEADLINE1		290.08 (317.26)	
NEW_DEADLINE2		151.10 (105.54)	
NEW_DEADLINE_ABOVE		94.68** (38.56)	
SLOTPICK_2003	-197.29*** (38.64)	-191.73*** (40.47)	-191.50*** (40.09)
SLOTPICK_2004	-157.59*** (27.29)	-151.60*** (26.07)	-150.75*** (25.85)
SLOTPICK_2005	-150.05*** (31.46)	-159.73*** (28.29)	-159.40*** (27.94)
SLOTPICK_2006	-188.02*** (40.04)	-183.28*** (36.78)	-182.65*** (36.78)
SLOTPICK_2007	-331.83*** (70.91)	-323.88*** (51.00)	-322.96*** (51.24)
SLOTPICK_2008	-195.58*** (63.15)	-173.41** (53.41)	-172.82** (53.38)

(continued)

Appendix A

Table A I (continued)

Model	DV = Bonus or Guaranteed Compensation if Major League Contract	Pooled OLS (Value) (2003-09)	Pooled OLS (Revenue) (2003-09)
SLOT PICK_2009	-175.86*** (49.69)	-178.87*** (48.70)	-193.93*** (48.72)
SLOT PICK_2010	-181.54*** (61.02)	-184.66*** (60.85)	-175.96*** (53.18)
YEAR = 2004	-13.93 (11.54)	-14.11 (11.62)	-17.52 (11.44)
YEAR = 2005	19.99* (11.72)	19.78 (11.90)	11.29 (12.72)
YEAR = 2006	20.42* (10.81)	20.14* (11.12)	4.03 (14.39)
YEAR = 2007	-38.58 (42.11)	-25.09 (30.93)	-55.47 (36.60)
YEAR = 2008	-31.11 (45.89)	-16.38 (35.59)	-54.08 (37.89)
YEAR = 2009	-22.78 (43.62)	-8.79 (34.50)	-45.24 (40.75)
YEAR = 2010	-5.66 (42.99)	7.47 (33.47)	-32.57 (37.20)
VALUE		0.096*** (0.035)	
REVENUE			0.430** (0.183)
Observations	2,260	2,260	2,260
R ²	0.82	0.82	0.83

Authors' Note

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Notes

1. Also referred to as the "Rule 4 Draft" or the "Amateur Draft."
2. Wylde, J. and Wilson, S. (2007), "Origin of MLB Players," http://www.capecodbaseball.org/News/news2007/images/MLB_Players_2001_2006.pdf (accessed on February 13, 2008).
3. Unless the player explicitly waives this restriction in writing.
4. Technically, agents can only be used as advisors to the player and the player's family and cannot negotiate directly with the team, if the player wishes to maintain his college eligibility until a contract is signed. However, this rule is rarely enforced in baseball and agents often negotiate directly with the team without the player losing his college eligibility. One recent court ruling affirmed a baseball player's right to be represented by an agent in contract negotiations in one of the few cases of agent representation contested by the NCAA. See "Oliver Wins Suit Against NCAA," *Baseball America College Blog*, February 12, 2009, <http://www.baseballamerica.com/blog/college/?p=746>
5. For more on the history of the bonus rule and the rule First Year Player, see Simpson (2005).
6. In rare cases, players are drafted out of the independent minor leagues.
7. A college senior could re-enter the draft in the future if he does not sign. However, without the ability to play amateur baseball at a competitive level—and with an extra year of age—his perceived value would depreciate rapidly. For this reason, almost all drafted seniors sign a contract.
8. Before 2008, a fifth-year senior (e.g., a senior who "redshirted" 1 year because of injury) on a college team whose season had ended before the draft could be signed as a free agent before the draft. This rule was changed for the 2008 draft so that fifth-year seniors can no longer sign as free agents before the draft. See <http://www.baseballamerica.com/blog/draft/?p=272> (Accessed on May 25, 2011).
9. In 2003, the type of 5 players could not be determined. In 2004, the type of 1 player could not be determined. In 2005 and 2009, 2 players each year were drafted from the

independent leagues. In 2006, 2007, and 2010 one player each year was drafted from the independent leagues.

10. The first-round bonus averages from 1965-2002 were taken from "Bonus Record: Year-by-Year Progression" Baseball America. <http://www.baseballamerica.com/today/2005draft/050604bonus2.html> (Accessed on May 25, 2011).
11. In exchange for these concessions, the player's union negotiated an elimination of the penalty for signing mid-tier (denoted as "Type B") free agents and a narrower definition of top-tier ("Type A") and mid-tier free agents. When a team loses a player to another team through free agency, the team may receive compensation (in the form of draft picks) depending on the classification of the player lost. In the old CBA, if a team lost a Type A or Type B free agent, it would receive a high-round draft pick from the acquiring team and a draft pick in the first supplemental round in the subsequent draft. The draft pick lost by the acquiring team effectively acted as a tax on free agent signings, reducing the amount teams were willing to pay for free agents. In the new CBA, a team signing a Type B free agent no longer loses a draft pick, while the team that lost the free agent only gains a supplemental round draft pick. Thus, there is no longer a disincentive, in the form of lost draft picks, to sign Type B free agents. Signing Type A free agents still involves a transfer of a high-round draft pick from the acquiring team to the losing team, but the definition of a Type A free agent was narrowed to the top 20% of players, down from the top 30%. The definition of a Type B free agent was also changed to the players in the 21–40 percentile range instead of those between 31% and 50%. Thus, the penalty for signing above-average free agents, apart from those in the top 20%, was eliminated, presumably leading to higher salaries for these good, but not great, players. See "New Agreement Includes Draft, Rule 5 Changes," Baseball America, <http://www.baseballamerica.com/today/draft/news/262720.html> (Accessed on May 25, 2011) and "MLB, MLBPA Reach Five-Year Labor Accord," http://mlb.mlb.com/news/press_releases/press_release.jsp?ymd=20061024&content_id=1722380&vkey=pr_mlb&fext=.jsp&c_id=mlb (Accessed on September 2, 2011).
12. The negotiation deadline was extended to August 17 in 2009 because August 15, 2009, fell on a Saturday. In 2010, the negotiation deadline was extended to August 16 because August 15, 2010, fell on a Sunday.
13. In 2009, it was determined that players drafted out of the independent leagues, like seniors, are also not subject to the August 15 signing deadline. See "Clock Ticking for Teams to Sign Top Picks," http://mlb.mlb.com/news/article.jsp?ymd=20090804&content_id=6245030&vkey=news_mlb&fext=.jsp&c_id=mlb (accessed on May 25, 2011)
14. Baseball America 2009 Almanac, page 503.
15. Callis, Jim, "MLB Tries Again to Deflate Bonuses," Baseball America, June 15, 2009 (#0913), p. 16.
16. Harsanyi and Selten (1972).
17. Kahneman and Tversky (1979).
18. For a review of these studies, see Szymanski (2003).
19. Wins above replacement (WAR) is defined as the number of wins generated by a player above that which would have been generated by a replacement-level player (e.g., AAA or

- bench player), taking into account hitting, fielding, and pitching contributions. See <http://www.fangraphs.com/library/index.php/misc/war/> for a discussion of the techniques used to calculate WAR.
20. An alternate specification, with results reported in the Appendix A, includes the signing bonus or the total guaranteed compensation, if the player signed a major league contract, as the dependent variable.
 21. Links to all of Baseball America's draft resources, including the advanced draft database (ADD) for each year, can be found at <http://www.baseballamerica.com/today/draft/draftarchive.html>. Most of these resources, including the advanced draft database for each year, are only available with a subscription to Baseball America.
 22. Minor league contracts can also include nonpecuniary and semipecuniary compensation such as tuition assistance. The nonbonus items of the contract are not described in the ADD.
 23. <http://www.thebaseballcube.com/draft/index.shtml>
 24. <http://www.baseballamerica.com/online/draft/news/264528.html>
 25. All dollar values are nominal.
 26. The estimated type coefficients (HS, CCSO, and SR) may include valuation differences associated with the mean perceived differences in risk and proximity to the majors between high school players and college players, to the extent that these differences are not captured by the estimated coefficients of SLIDE and the PICK polynomial. See Spurr (2000) and Burger and Walters (2009) for a description of these valuation differences.
 27. The dependent variable, BONUS, is denominated in thousands of dollars. Clustered (on team) robust standard errors are in parentheses. The coefficients are statistically significant at the 99% (***), 95% (**), and 90% (*) levels as indicated. The p value for the F test of joint significance for the PICK polynomial terms is less than .001 in all cases.
 28. COMP1_NEW and NEW_DEADLINE1 are jointly significant at the 99% level. Likewise, COMP2 and NEW_DEADLINE2 are jointly significant at the 95% level.
 29. Storen's pre-draft rank according to Baseball America was #36.
 30. Using the primary bonus model.
 31. Also excluding players drafted from the independent leagues.
 32. The variables that track the changes from the new CBA (COMP1_NEW, COMP2, COMP3, NEW_DEADLINE) are jointly significant at the 95% level.
 33. Callis, J., "MLB Fails to Recognize Draft Truths," *Baseball America*, September 5, 2011, p. 11
 34. Ringolsby, T., "After Years of Tweaks, Baseball Contemplates Big Draft Changes," *Baseball America*, June 13, 2011, pp. 12–13.
 35. The difference between the SLOTPICK_2006 and SLOTPICK_2007 coefficients is statistically significant at the 95% level.
 36. The difference between the coefficients for SLOTPICK_2007 and SLOTPICK_2009 is statistically significant at the 95% level.
 37. Does not include contracts that were signed and then voided. The second round includes the first and second supplemental rounds. The third round includes the third supplemental round.

38. At the means of the independent variables.
39. Standard errors of the marginal effects are in parentheses. The coefficients (not necessarily the marginal effects) are statistically significant at the 99% (***), 95% (**), and 90% (*) levels as indicated.

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